Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/13/2013
PI Name:	Yu, Nan Ph.D.		
Project Title:	JPL Participation in ESA ACES (Atomic Clock Ensemble in Space) Worldwide Clock Comparison Campaign		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	09/30/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	In additional to PI/Co-I, the following people are making contributions to the task in FY13: William Diener, Fang Zhong, Wade Rellergert, and Lukas Baumgartel.		
COI Name (Institution):	Burt, Eric (Jet Propulsion Laboratory) Prestage, John (Jet Propulsion Laboratory) Tjoelker, Robert (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15743		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tremendous progress has been made recently in the U.S. and other countries in advancing atomic clock technology and its applications for the benefit of science, national interests, and society in general. Space provides a unique environment in which many benefits can be realized. The European Space Agency (ESA) is readying the ACES (Atomic Clock Ensemble in Space) payload for the International Space Station (ISS) for an anticipated 2013 launch. Some of the key objectives of ACES will be achieved by global time transfer and clock comparisons with the most advanced clocks at multiple worldwide sites. There is an opportunity for investigators from the leading U.S. clock and metrology labs to participate in the clock comparison campaign, enhancing the overall science objective of ACES and at the same time demonstrating and providing science benefit from advanced U.S. clock technology. JPL is one of the major U.S. time and frequency metrology institutions serving NASA missions and Deep Space Network needs. It was the NASA center for the microgravity LCAP program (Laser Cooling and Atomic Physics) in charge of developing the cold atom clock payload for PARCS (Primary Atomic Reference Clock in Space). JPL has also developed a unique atomic frequency standard that has demonstrated an exceptional long-term stability. We propose to collaborate with ESA on the ACES mission by participating in the ACES global clock campaign with JPL's ultra-stable mercury ion clock. During a proposed six-year performance period, we will: 1) contribute significantly to the ACES mission science objectives by providing unique clock references from a Western U.S. site; 2) Provide and demonstrate a high-impact application for JPL's high performance time keeping clock technology; and 3) re-establish NASA's involvement in ISS Clock Science and initiate a healthy collaboration between NASA and ESA in the area of clock and quantum sensors. The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal provided by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is only ground comparison site located in the western USA and the mercury ion clock is the only one of its kind in the global clock comparison campaign, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advanced atomic clocks are critical to the interests of NASA and the nation. Telecommunication and navigation (e.g., GPS) are well-known applications of space clocks. Within NASA, high performance atomic clocks and frequency and timing capabilities enable operation of the NASA Deep Space Network and provide fundamental tools for precise measurements exploring the laws of Nature. Progress in advanced clock technology development, both on ground and for space, is rapidly being made in many countries. Participation by the U.S. in the ACES mission contributes to the science objectives of the ISS consistent with the 2011 NRC report recommendations in Fundamental physics and demonstrates the capability and potential of NASA mercury ion clock technology.
Task Progress:	This is the third year of the multi-year task. The first year involved formally establishing the collaboration, beginning needed updates of a JPL ground ion clock and preparation for the installation of the ESA provided microwave link ground terminal at a JPL site. The second year focused on LITS-9 upgrade, its diagnosis and the design of the MWL ground terminal site. This year, the ground clock development focus has been shifted to leveraging the new microwave ground clock development efforts. These ground clocks are the next generation LITS microwave clocks based trapped mercury ions. The progress of these LITS clock development is on target. Initial test results show clock characteristics as expected. One of these new LITS locks will be available for ACES project together with LITS9 and the whole ensemble of high performance clocks at JPL. At the same time, an optical clock based on a single trapped ion is being developed at JPL that can be leveraged by ACES and is expected completed in time for comparison. The impact of a high performance optical clock in the ACES clock comparison at JPL has been investigated. In addition, the preparation for the microwave link (MWL) ground terminal has been moving along. We have done more analysis of ISS viewing at possible JPL sites and have completed the formal design of the site for MWL terminal. The construction of the ground terminal site has mostly been completed this fiscal year. Furthermore, we have also studied and simulated the actual ISS viewing conditions at JPL and MWL terminal link times. Using the expected performance of the JPL clock stability and ISS orbit parameters, we simulated the clock comparison and analyzed some of the related science implications.
Bibliography Type:	Description: (Last Updated: 10/28/2020)
Abstracts for Journals and Proceedings	Tjoelker RL, Burt EA, Diener W, Enzer D, Prestage JD, Yu N. "JPL Activities in Preparation for the ACES Mission. " 2012 European Frequency and Time Forum, Gothenburg, Sweden, April 23-27, 2012. Poster presentation. 2012 European Frequency and Time Forum. Proceedings of a meeting held 23-27 April 2012, Gothenburg, Sweden. Printed by Curran Associates, Inc. 2012. , Apr-2012
Abstracts for Journals and Proceedings	Tjoelker RL, Burt EA, Prestage JD. "High Stability Mercury Ion Clocks for Science, Navigation, and Exploration Applications." 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. , Jun-2013
Abstracts for Journals and Proceedings	Yu N. "ISS Fundamental Physics Activities at JPL." 6th From Quantum to Cosmos Workshop, Nice, France, October 15-17, 2013. 6th From Quantum to Cosmos Workshop, Nice, France, October 15-17, 2013. , Oct-2013
Abstracts for Journals and Proceedings	Yu N, et al. "JPL ACES Collaboration - Fundamental physics measurements with precision clocks aboard the ISS and on the ground." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts. , Nov-2013